A5 Asynchronous Assignment 5: Electrical Protection

Sunday, October 18, 2026 2:46 PM

Comprehensive Circuit Protection Design and Component Selection

Objective

Design robust protection mechanisms for power and data lines, select viable commercial components, and justify your engineering choices in a formal report.

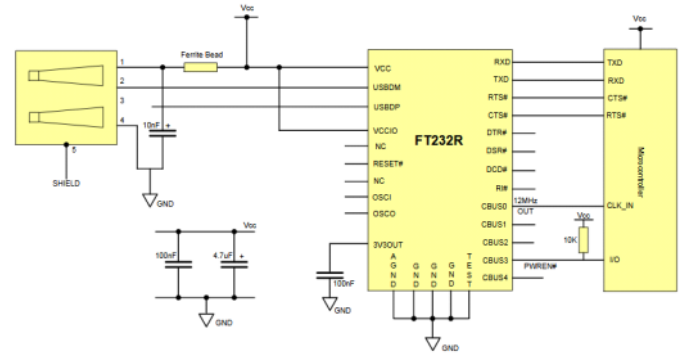
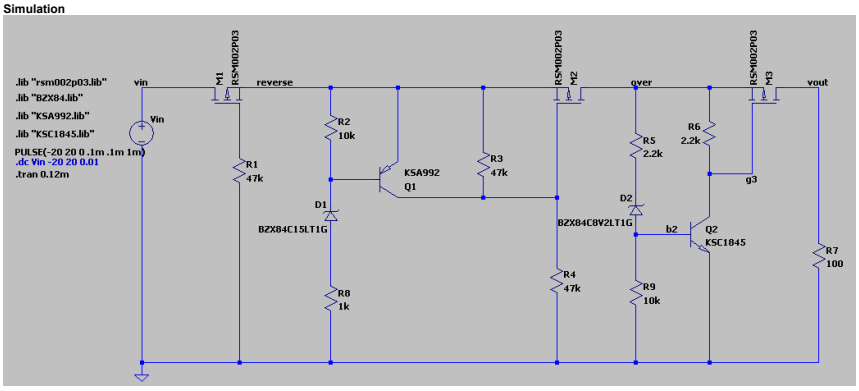


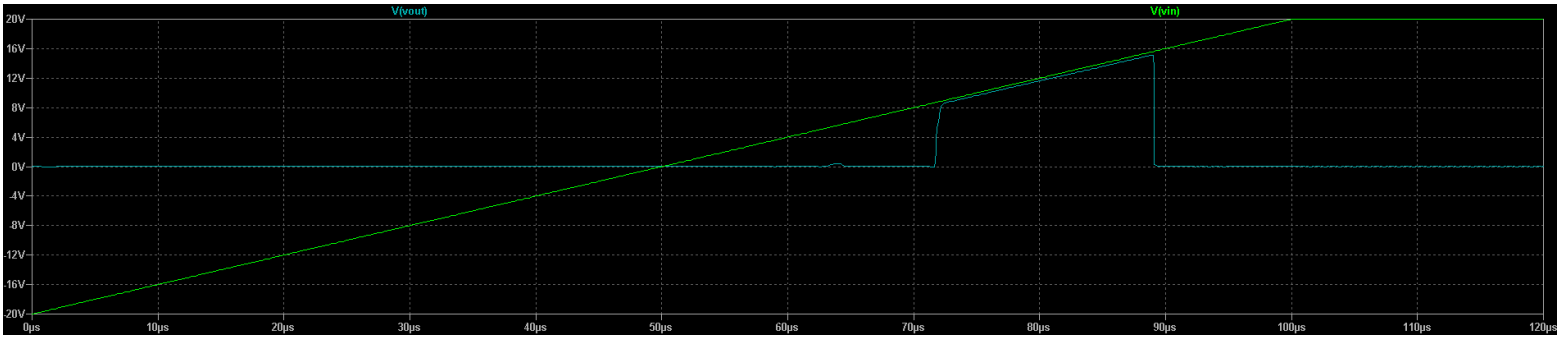
Figure 7.4 USB to MCU UART Interface

1. Active Power Line Protection Design two active protection circuits for a typical automotive application with an operational range of 9-16V:

Overvoltage Protection: Design a Zener-based circuit that disconnects the load when the input voltage exceeds 16V.
Undervoltage Lockout: Design a complementary circuit that turns off the load when the input voltage drops below 9V.
Reverse polarity protection: Make sure there is a mechanism that disconnects the circuit if the polarity was connected inverse.

Simulation





Part numbers

MOSFET transistors

Número de parte del fabricante	Cantidad disponible	Tipo FET	Voltaje de drenaje a fuente (Vdss)	Vgs (máx.)	Voltaje de impulso (encendido Rds máx, encendido Rds mín.)	Corriente - consumo continuo (Id) a 25°C
RSM002P03T2L MOSFET P-CH 30V 200MA VMT3 Rohm Semiconductor	14,308 En stock	Canal P	30 V	±20V	4V, 10V	200mA (Ta)

BJT control transistors

Número de parte del fabricante	Cantidad disponible	Tipo de transistor	Corriente - Colector (Ic) (Máx.)	Voltaje - Ruptura de emisor colector (Máx.)	Precio Precio por cantidad	Serie	Paquete	Estado del producto	Vce de saturación (máx.) a Ib, Ic	Corriente - Corte de colect
KSA992FBTA TRANS PNP 120V 0.05A TO-92-3 Fairchild Semiconductor	5,670,100 Mercado	PNP	50 mA	120 V	4,494 : $0.07000 Granel	-	Granel	Activo	300mV a 1mA, 10mA	1µA

Número de parte del fabricante	Cantidad disponible	Tipo de transistor	Corriente - Colector (Ic) (Máx.)	Voltaje - Ruptura de emisor colector (Máx.)	Precio Precio por cantidad	Serie	Paquete	Estado del producto	Vce de saturación (máx.) a Ib, Ic	Corriente - Corte de colect
KSC1845FTA TRANS NPN 120V 0.05A TO-92-3 onsemi	87,920 En stock Este producto tiene un límite de compra máximo.	NPN	50 mA	120 V	1 : $0.33000 Cinta cortada (CT) 2,000 : $0.07388 Cinta y caja	-	Cinta cortada (CT) Cinta y caja	Activo	300mV a 1mA, 10mA	50 nA (ICBO)

Zener diodes for clamping 15V and 8.2V

Número de parte del fabricante	Cantidad disponible	Precio (Precio por cantidad)	Serie	Paquete	Estado del producto	Voltaje - Zener (Nom) (Vz)	Tolerancia	Potencia - Máx.	Impedancia (Máx.) (Zzt)	Corriente - Fuga inversa a Vr	Voltaje - Directo (Vf) (m
BZX84C15LT1G DIODE ZENER 15V 250MW SOT23-3 onsemi	366,783 En stock	1 : $0.16000 Cinta cortada (CT) 3,000 : $0.03125 Cinta y rollo (TR)	BZX84CxxxLT1G	Cinta y rollo (TR) Cinta cortada (CT) Digi-Reel®	Activo	15 V	±6%	250 mW	30 Ohms	50 nA @ 10.5 V	900 mV @ 10 mA

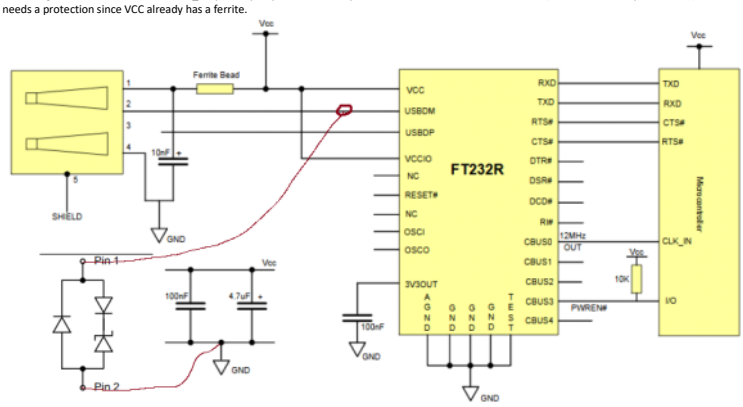
Número de parte del fabricante	Cantidad disponible	Precio (Precio por cantidad)	Serie	Paquete	Estado del producto	Voltaje - Zener (Nom) (Vz)	Tolerancia	Potencia - Máx.	Impedancia (Máx.) (Zzt)	Corriente - Fuga inversa a Vr	Voltaje - Directo (Vf) (máx.) a
BZX84C8V2LT1G DIODE ZENER 8.2V 250MW SOT23-3 onsemi	0 En stock 780,391 Mercado Consultar el plazo de entrega	1 : $0.16000 Cinta cortada (CT) 3,000 : $0.03125 Cinta y rollo (TR) 6,575 : $0.05000 Granel	BZX84CxxxLT1G	Cinta y rollo (TR) Cinta cortada (CT) Digi-Reel® Granel	Activo	8.2 V	±5%	250 mW	15 Ohms	700 nA @ 5 V	900 mV @ 10 mA

Part 1 design conclusions.
The design of this protection circuits was done with the help of AI which help me seeing more scenarios that were not conside red in my initial solution, for example the current limitation in the D1 with R8, "absorbing" the leakage current of D2 with R9, or reducing the gate response of M3 with a lower R6. Additiona l scenarios where inputs are missing and can affect the selection of the components like the temperature at which the example circuit is operating or the maximum surge value. The ex ercise included the use of transistors which I was not familiar to work with, at least from a design point of view. IA also helped me with how they are polarized. Finally as part of simulat ion I have included the actual part numbers using the SPICE models from the vendors.

Part 2: USB Data Line Protection (TVS) Based on the provided application notes for an FT232R USB-to-serial converter (specifically referencing the suggested application in Figure 7.4):

Select appropriate TVS (Transient Voltage Suppression) components to protect the signals exiting the board.
Analyze the interface and answer: Do all connector signals require protection?

According to the TVS Quick Start Guide_May 2012.pdf, protection is required in all data lines D1+, D1- and VCC (that are externally connected). For the example circuit the USBDM is the only one line that needs a protection since VCC already has a ferrite.



Using a single line protection from the recommendations in TVS Quick Start Guide_May 2012.pdf.

Número de parte del fabricante	Cantidad disponible	Precio (Precio por cantidad)	Serie	Paquete	Estado del producto	Tipo	Canales unidireccionales	Voltaje - Separador invertido (Típico)	Voltaje - ruptura (mín.)	Voltaje - sujeción (máx.) a Ipp	Corriente - pul
ESD5V3U1U02LRHE6327XTSA1 TVS DIODE 5.3VWM 28V PGTSLP27 Infineon Technologies	610,745 Mercado	2,834 : $0.11000 Granel	-	Granel	Activo	Zéner	1	5.3V (máx.)	6V	28V (típico)	3 A (8/20µs)